ISAAC PANDRAJU

Senior Data Engineer | Azure • Databricks • PySpark • GenAI Data Platforms
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PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years building production data platforms on **Azure Databricks, PySpark, and Delta Lake**. Experience spanning **GenAI/RAG data pipelines**, **Medallion (Bronze/Silver/Gold) architecture**, dimensional modeling, and **CI/CD**-driven deployment across manufacturing, F&B, aviation, FMCG, and financial clients. Microsoft Certified Azure Data Engineer and 5x Databricks certified. Led the intern-to-FTE technical pipeline for 30+ engineers.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PySpark, Linux Shell Scripting
- **Data Engineering:** ETL/ELT, Medallion Architecture, Data Warehousing, Dimensional Modeling (Star Schema, Fact/Dimension), Slowly Changing Dimensions (SCD Type 1/2), Data Modeling, Data Quality, Data Governance, Incremental & CDC Loads, Performance Tuning
- **Distributed Computing:** Apache Spark, Spark SQL, Structured Streaming, Apache Hadoop, Apache Hive
- **Databases & Warehouses:** Azure Synapse Analytics, PostgreSQL, MySQL, Azure Cosmos DB (NoSQL), Delta Lake
- **Azure Services:** Azure Databricks, Azure Data Factory, Azure Data Lake Storage (ADLS Gen2), Azure Blob Storage, Azure Synapse Analytics, Azure Key Vault, Azure Event Hubs, Azure Service Bus, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Azure App Service
- **GenAI / RAG:** Retrieval-Augmented Generation, Vector Search & HNSW Indexing, Embeddings (text-embedding-3-large), Token-Aware Chunking (tiktoken), Citation Mapping
- **DevOps & Tooling:** Azure DevOps, Git, CI/CD, Databricks Asset Bundles (DABs), Databricks Workflows, Unity Catalog, pytest, YAML, Python Packaging (wheel/hatchling), Kafka (basics)
- **Cloud:** Azure (primary), AWS

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- Databricks Certified Data Engineer Professional
- Databricks Certified Data Engineer Associate
- Databricks Certified Associate Developer for Apache Spark 3.0
- Databricks Certified Data Analyst Associate
- Databricks Certified Generative AI Engineer Associate

PROFESSIONAL EXPERIENCE

Celebal Technologies — *Sep 2022 – Present*

Senior Software Engineer — *Hyderabad, India*

Industrial Equipment Manufacturer (Mining / Construction) — GenAI / RAG Data Platform — *Mar 2025 – Present*

Tech stack: Azure Databricks, PySpark, Delta Lake, Unity Catalog, Databricks Asset Bundles, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Cosmos DB, ADLS Gen2, Azure DevOps, Python, SQL

- Architected and delivered a **config-driven data platform** powering **2 production RAG chatbots** over industrial equipment manuals, enabling natural-language Q&A with **exact PDF page citations** for field technicians across **6 document divisions**.
- **Re-engineered a duplicated legacy codebase into a single versioned Python wheel**, eliminating 3–5x transform duplication per division by driving all division differences (table names, composite keys, feature flags, index targets) from **YAML descriptors** instead of copied notebooks — reducing new-division onboarding from a full notebook rewrite to one config file.
- Built end-to-end **Medallion (Bronze/Silver/Gold) pipelines** on **Delta Lake** covering ZIP ingestion, HTML/XML/PDF parsing, metadata extraction, entity mapping, token-aware chunking, embedding generation, and vector indexing.
- Engineered **distributed document processing with PySpark mapInPandas**, parallelizing extraction of **800 MB+ ZIP packages** and BeautifulSoup HTML crawls across executors — replacing single-driver processing and removing the primary pipeline bottleneck.
- Designed a **4-stage page-mapping waterfall** (Exact → Suffix → Fuzzy → Chapter) joining extracted HTML content to PDF table-of-contents bookmarks, achieving near-complete content-to-source-page mapping and enabling **verifiable citations** in chatbot answers.
- Developed **Azure OpenAI embedding pipelines** (text-embedding-3-large, 3072-D) with batched requests and retry/backoff, and built **HNSW vector indexes in Azure AI Search** with idempotent upsert-first pushes at 500-doc batches across 16 partitions.
- Built an **incremental orchestration layer** — change detection classifying source packages as NEW/UPDATED/UNCHANGED, a **Delta audit table**, per-entity **resume after failure**, soft-delete of stale Silver/Gold rows, and market-hold gating — cutting reprocessing to only changed documents.

- Orchestrated production runs as **14-task Databricks Workflow DAGs** (one per division) with fan-in dependencies, task-value hand-off between stages, per-task retries, and **repair-run recovery** from the failed task.
- Implemented **CI/CD via Azure DevOps multi-stage pipelines** — parallel pytest and bundle validation across both targets, auto-deploy to dev, **approval-gated** deploy to test, service-principal authentication, and **Unity Catalog** grants.
- Hardened security and reliability: migrated hard-coded plaintext credentials into **Azure Key Vault-backed secret scopes** with per-environment key templating, enforced **RBAC** and structured logging, and authored a **10-module pytest suite** including a build-breaking guard against hard-coded catalog literals.
- Developed a **Cosmos DB → Delta feedback ingestion pipeline** to capture user interactions and response ratings, creating the analytics loop used to measure and improve chatbot answer quality.

Food & Beverage Sector — Data Engineer *Apr 2024 – Feb 2025*

Tech stack: PySpark, SQL, Azure Databricks, Delta Lake, Databricks Workflows, Azure Data Factory, Azure DevOps, Data Modeling

- Ingested and integrated diverse source datasets — **sales, customer, promotion, and ML model outputs** — through a **Medallion architecture**, delivering clean, scalable, centrally governed data for analytics.
- Modeled curated **Gold-layer dimensional datasets** enabling business stakeholders to track **promotional effectiveness, customer trends, and revenue performance** in near real-time.
- Implemented **CI/CD workflows in Azure DevOps** to deploy, monitor, and version-control pipelines, **reducing deployment time and operational overhead by 50%**.
- Delivered promotional-effectiveness insights that enabled targeted campaigns and measurable uplift in tracked sales performance.

Aviation Sector — Data Engineer *Jul 2023 – Mar 2024*

Tech stack: PySpark, Python, Azure Databricks, Delta Lake, ADLS Gen2, Azure Data Factory

- Built scalable **ETL pipelines in Azure Data Factory**, **reducing batch processing time by 30%**.
- Engineered explicit schemas for **fixed-width source files**, improving ETL reliability and downstream data quality.
- Developed a **Medallion architecture** on Databricks and **Delta Lake** to standardize and streamline data transformation.
- Transformed complex **nested JSON** payloads into optimized, query-efficient Delta tables using **PySpark**.

FMCG Sector — Data Engineer *Sep 2022 – Jun 2023*

Tech stack: Python, SQL, PySpark, Azure Databricks, Azure Data Factory, SCD, SAP BW

- Migrated data from heterogeneous source systems to **Azure Data Lake Storage** using **Azure Data Factory**, improving end-to-end processing time.
- Designed and deployed a reusable, **metadata-driven ingestion framework** in Azure, improving data flow efficiency across source systems.
- Authored a Qlik decomposition document detailing data mappings and transformation logic, ensuring seamless integration with **SAP Business Warehouse**.
- Implemented **SCD Type 1** logic for accurate dimension updates and historical tracking.
- Conducted extensive data quality validation using **Spark SQL** and Excel, achieving high accuracy in data issue resolution.

Futurense Technologies *Nov 2021 – Sep 2022*

Data Engineer *Bangalore, India*

Financial Sector — Data Engineer *Nov 2021 – Sep 2022*

Tech stack: SQL, PySpark, Azure Databricks, Delta Lake

- Optimized existing **Azure Databricks** notebooks to **reduce pipeline runtimes by 40%**, improving processing efficiency and overall throughput.
- Improved storage and retrieval performance by implementing **Z-Ordering** on Delta tables, **boosting query performance by 30%** for large-scale analytics workloads.
- Enhanced ETL performance through **Spark configuration tuning, partitioning strategy redesign, and caching**, reducing cluster cost and job latency.

EDUCATION

NRI Institute of Technology (NRIIT) *Jun 2018 – Apr 2022*

B.Tech., Electronics and Communication Engineering | **CGPA: 9.2/10**

LEADERSHIP & AWARDS

- **Special Appreciation Award** — for creating and delivering **Databricks Performance Tuning and Optimization** training as a contribution to the Catalyst program.
- Led the end-to-end **intern-to-FTE pipeline for 30+ candidates**, managing training, technical mentoring, interview evaluation, and onboarding into full-time roles.
- Designed and delivered structured training programs on **Azure Databricks, PySpark, and ETL development**, improving onboarding readiness and code quality across incoming engineers.